

Batch Code: ANP_D1838

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LAB-06

Questions:

1. Write a java program using Multi-Level Inheritance and explanation
 2. Write a java program using Interface [Multiple inheritance] and explain why java doesn't support multiple inheritance
-

Program:

1. Multi-Level Inheritance:

```
class Son {
    int a;

    public Son(int a) {
        this.a = a;
    }
}
class Father extends Son {
    int b;

    public Father(int a, int b) {
        super(a);
        this.b = b;
    }
}
class GrandFather extends Father {
    int c;

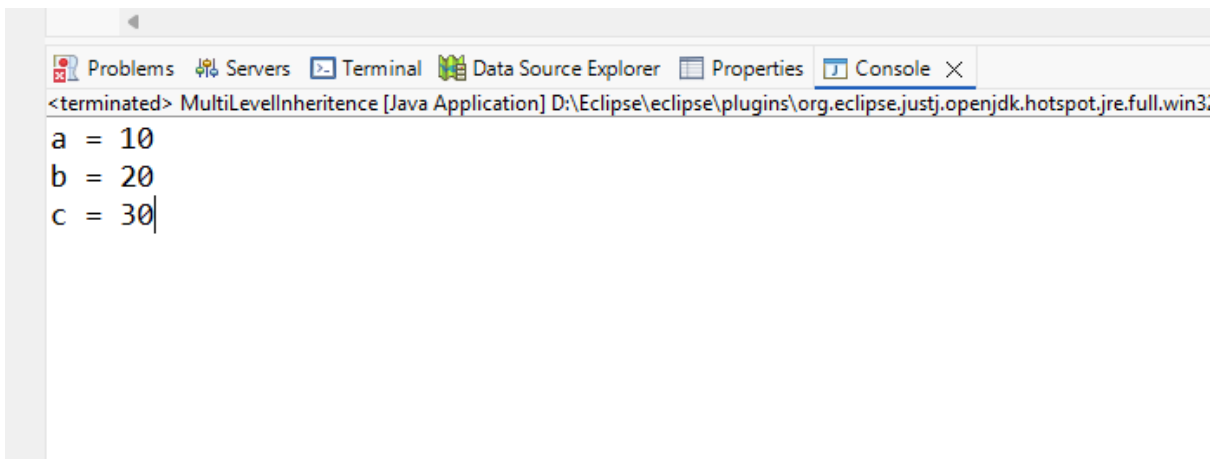
    public GrandFather(int a, int b, int c) {
        super(a, b);
        this.c = c;
    }
}
```

```

    }
    public void display() {
        System.out.println("a = " + a);
        System.out.println("b = " + b);
        System.out.println("c = " + c);
    }
}
public class MultiLevelInheritance {
    public static void main(String[] args) {
        GrandFather obj = new GrandFather(10, 20, 30);
        obj.display();
    }
}

```

Output:



```

<terminated> MultiLevelInheritance [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32
a = 10
b = 20
c = 30

```

- class Son
 Defines a class named Son.
 It has one variable a.
 Constructor Son(int a) initializes that variable.
- class Father extends Son
 Defines a class named Father.
 extends Son → Father inherits all properties of Son.
 It adds a new variable b.
 Constructor Father(int a, int b) calls super(a) to initialize a from Son, and

sets b.

- class GrandFather extends Father

Defines a class named GrandFather.

extends Father - GrandFather inherits everything from Father (and indirectly from Son).

Adds a new variable c.

Constructor GrandFather(int a, int b, int c) calls super(a, b) to initialize a and b, then sets c.

display() method prints values of a, b, and c.

- public class MultiLevelInheritance

Main class with main() method.

Creates an object obj of type GrandFather.

Passes values (10, 20, 30) to constructor.

Calls obj.display() - prints all three values.

2. Multiple Inheritance:

```
interface Java {
```

```
    void javaProgram();
```

```
}
```

```
interface Python {
```

```
    void pythonProgram();
```

```
}
```

```
interface C {
```

```
    void cProgram();
```

```
}
```

```
public class MultipleInher implements Java, Python, C
```

```
{
```

```

public void javaProgram()
{
    System.out.println("Java Programming");
}

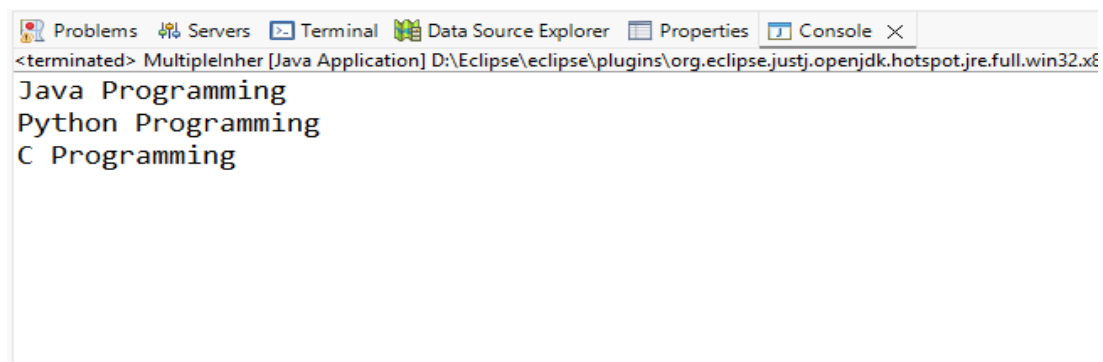
public void pythonProgram()
{
    System.out.println("Python Programming");
}

public void cProgram()
{
    System.out.println("C Programming");
}

public static void main(String[] args)
{
    MultipleInher obj = new MultipleInher();
    obj.javaProgram();
    obj.pythonProgram();
    obj.cProgram();
}
}

```

Output:



The screenshot shows the Eclipse IDE's Console window. The title bar includes tabs for Problems, Servers, Terminal, Data Source Explorer, Properties, and Console. The console output displays the results of the program execution: "Java Programming", "Python Programming", and "C Programming" on separate lines. The window title is "<terminated> MultipleInher [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64.jdk\bin\java.exe".

```

<terminated> MultipleInher [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64.jdk\bin\java.exe
Java Programming
Python Programming
C Programming

```

- interface Java
 - Defines an interface named Java.
 - Declares a method javaProgram() (no body, just a contract).
- interface Python
 - Defines an interface named Python.
 - Declares a method pythonProgram().
- interface C Defines an interface named C.
 - Declares a method cProgram().
- public class MultipleInher implements Java, Python, C
 - Defines a class named MultipleInher.
 - implements Java, Python, C - this class promises to provide code for all methods declared in those interfaces.
- Method Implementations
 - public void javaProgram() - prints "Java Programming".
 - public void pythonProgram() - prints "Python Programming".
 - public void cProgram() - prints "C Programming".
- public static void main(String[] args)
 - Entry point of the program.
 - Creates an object obj of type MultipleInher.
 - Calls all three methods - executes the print statements.

Why java does not Support Multiple Inheritance?

- Java does not support multiple inheritance of classes because it can create confusion when two parent classes have methods with the same name.
- This situation is called the Diamond Problem.
- To keep the language simple and avoid ambiguity, Java only allows single inheritance.

- However, Java provides multiple inheritance through interfaces, which lets a class inherit behaviour from many sources safely without conflict.
- Another reason is that single inheritance improves readability and reduces errors, making Java programs more reliable.

